Matthew Morgan

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Education

Iowa State University Ames, Iowa — Expected Graduation: May 2027

Bachelor of Science, **Software Engineering** — GPA 3.99/4.00

Work Experience

Apple, San Diego, CA — May 2026 – August 2026

Software Engineering Intern – Posters Team

- Designed an **extension point** allowing independently-owned processes to contribute data to a host process: a public protocol, host-side plumbing, a reference implementation, unit test suites for both new components, and an adopter-facing integration guide
- Prototyped a **new user flow** for interacting with Wallpapers
- **Drove adoption** in another team's codebase, including a background job to **migrate data** for existing installs.
- Surfaced ~20 tuning parameters through an internal live-configuration tool so designers could iterate on-device in real time without a rebuild,
- Extended the tuning tool itself, which is used across the organization, to **render inline documentation** per parameter, making the controls self-describing.

Ecolab, Naperville, IL

Innovation Engineer — August 2024 – May 2026

- Rapidly developed **revenue generating prototypes** by leveraging our data to solve customer problems.
- Developed a novel approach to leveraging free text data across multiple data sources for customer success
- Served as a **flexible technical contributor**, working on high-impact projects wherever additional resources or expertise was most needed.
- Continued to develop AI prototypes, internal demos, and customer solutions for top level executives.

Artificial Intelligence Intern — May 2024 – August 2024

- Developed Generative AI prototype to live-demo for the largest Ecolab customer.
- Built a product that identifies **$X millions** of missing Total Value Delivered, where our customers are not utilizing our products and services to the proper extent, which leads to customer safety risks.
- Created a **rapid prototyping** AI environment so we could make constant iterations of the prototype as we worked towards the final product to be presented to the Board.
- Shadowed and **mentored** under the **Director of Artificial Intelligence**.
- Collaborated with a team of 3 to successfully deliver an AI Innovation for **Ecolab's Board of Directors**.

Technical Projects

Whiteboard Polar Plotter — December 2026 – February 2026

- Built a **two-stepper polar plotter** that draws on a 5×3 ft whiteboard; ESP32 firmware converts Cartesian paths to appropriate belt lengths in real time with a holding line **error < 0.5 mm.**
- Wrote an **image-to-toolpath pipeline** that turns any photo into marker strokes via crosshatching and tonal shading, then reorders paths to **cut pen-up travel 92%** and total draw time from **~180 min to ~15 min**.
- Shipped a web front end and job queue for **submitting drawings from anywhere**; jobs stream to the device over **MQTT** with buffering and mid-job resume so Wi-Fi drops don't ruin a drawing.

Robotic Arm with Computer Vision Control | Robotics/Computer Vision — March 2025 – September 2025

- Built a **closed-loop robotic arm** system following the Open-Source Arctos v2.9.7 platform.
- Created a React-based web interface to visualize camera input and control the robot wirelessly.
- Developed a **vision pipeline** in Python with **OpenCV** to track objects for autonomous manipulation.
- Enabled robot with a **homing sequence** that allows for consistent movement and positional awareness.

Skills

Python, Pandas, NumPy, OpenCV, TensorFlow, Scikit-Learn, Flask, Gen-AI, Java, C, C++, HTML, CSS, JavaScript, React.JS, TypeScript, Selenium, MySQL, Webscraping, LangChain, Objective C, Swift, SwiftUI, UIKit